

A. Follow the "Assessing Table Structure" procedure shown. List multivalued dependencies, functional dependencies, candidate keys, primary keys, and foreign keys. State your assumptions.

Functional Dependencies:

Shipment Number -> Shipper, Departure, Arrival

ItemID (should be specified but is not)->Item Name ,Price, InsuredValue

Store-> Salesperson, Country,

Phone-> Contact

Primary Keys:

Phone Number primary key for contact. ShipmentNumber is a primary key for shipment ItemID would be a primary key for items.

Candidate Keys:

Phone Number is a unique identifier because no phone numbers are the same, Shipment Number is an identifier assuming each number is different, if an Item is given an id can be unique to identify different items.

Multivalued Dependencies:

ShipmentNumber has a multivalued dependency on contents and items, as well as city and from

Foreign Keys:

Item this is a foreign key because item names can be used as an identifier for a shipment, Phone Number, can be used to identify the customer

Assumptions:

I assume that this table is not even in 1NF. There are many ways the data overlaps and creates dependencies. Specifically with the Items and contents. These two identities can cause shipments to be mixed up because they are inconsistent across the data. Some contents are not mentioned as an item. The same goes for Shipper and City, for city states this can be a problem, as well as cities who have the same name.

B. List questions you would ask Phillip to verify your assumptions.

What differentiates the items from the contents?

Does the from value and the City value overlap?

Is the insured value based on the price?

Why is departure inconsistent with order placement (2009-1999)?

Why are items in contents not listed in Items.

Why are items not identified by a unique ID

Why are stores not identified by a unique ID

C. Create tables as necessary to eliminate multivalued dependencies, if any.

Shipment(Shipment Number, ShipperCountry (instead of shipper), Departure, Arrival)

Items(ItemID, ItemName, Price)

ShipmentItem(ItemID, ShipmentNumber, InsuredValue)

Store(StoreID, Country, Salesperson)

Customer(Phone, Contact)

D. The relationship between shipment and item data could be inferred by matching values in the From cells to values in the City cells. Describe two problems with that strategy.

Country and City, it would be hard to match that up cleanly especially because countries have multiple cities.

Also we have data from countries whose cities are not listed and vice versa, For the philippines you have manilla but singapore is a city and a state, also if a city is not listed that can cause other problems

E. Describe a change to this spreadsheet that does express the shipment/item relationship.

ShipperCountry needs to be expressed rather than worrying about the city. You then could filter by stores in said country, nations could have cities by the same name like Athens in GA US vs Athens in Greece. If we filter items by the country you can filter by store, and that will eliminate any confusion.

F. Assume that Phillip wishes to create an updateable database from this data. Design tables you think are appropriate. State all referential integrity constraints.

Shipment(ShipmentNumber, ShipperCountry (instead of shipper), Departure, Arrival)

Items(ItemID, ItemName, Price)

ShipmentItem(*ItemID*, *ShipmentNumber*, InsuredValue)

Store(StoreID, Country, Salesperson)

Customer(Phone, Contact)

Referential Integrity Constraints:

ShipmentItem.ShipmentNumber must exist in Shipment.ShipmentNumber

ShipmentItem.ItemID must exist in Items.ItemID

Customer.Phone must exist to reference a customer

Shipment.Departure must be less than Shipment.Arrival

Store.StoreID must be different for every single store

G. Assume that Phillip wishes to create a read-only database from this data. Design tables you think are appropriate. State all referential integrity constraints.

Shipment(Shipment Number, ShipperCountry, Departure, Arrival, ItemName, Store, CustomerName, CustomerPhoneNumber)

Shipment.Departure must be less than Shipment.Arrival

This design makes one table with all the necessary data but is not normalized because you don't need to update anything. It just needs to be stated as is for it to be reviewed.

H. Do these data have the multivalued, multicolumn problem? If so, how will you deal with it?

I Would fix the item city/ shipper country problem. Have any items not listed from contents added in to the item table. Make sure the contents match the same names as Item. Also separate tables into 3NF format.

I. Do these data have the inconsistent data problem? If so, how will you deal with it?

Yes shipper country and Item city are inconsistent, contents and items are inconsistent. For me I would first separate all the data. Update every item listed in contents into the items table, set the price. Make sure that arrival dates are always more than the departure dates. Make sure the shipping numbers are in a consistent format for international and worldwide orders.

J. Do these data have a null value data problem? If so, how will you deal with it?

Yes Null values when it comes to arrival. There needs to be updated dates for those who are null, if there is no confirmed date I would put an date based on estimated arrival, or the average arrival time.

K. Do these data have the general-purpose remarks problem? If so, how will you deal with it?

Yes the contents attribute contains data from items while having a description. This can easily cause inconsistencies in the data. I would get rid of the contents attribute, and make sure all items listed in contents are identified in items. I would then create a table joining shipment and items to make a clear path for identifying items for shipments.